

FP1 Chapter 5 Summation of Series

Lecture 5 Summation of Series

(Summation of Series, Method of Differences, Sum to Infinity, Convergent, Divergent, Sum to Infinity)

Summation of Series

Let S_n be the sum of the first n terms of the series whose r th term is u_r .

Then $S_n = \sum_{r=1}^n u_r = u_1 + u_2 + \cdots + u_r + \cdots + u_n$.

Examples:

Properties of the Summation Notation:

Properties	Proof
$\sum_{n=1}^k (a_n + b_n) = \sum_{n=1}^k a_n + \sum_{n=1}^k b_n$	
$\sum_{n=1}^k c a_n = c \sum_{n=1}^k a_n,$ for any constant c.	
$\sum_{n=p}^r a_n = \sum_{n=p}^q a_n + \sum_{n=q+1}^r a_n,$ where $p < q < r$.	

Method of Differences

- The technique to sum a series is known as method of differences.
- In this case, we try to express a u_r into the form $f(r + 1) - f(r)$. This allows $\sum_{r=1}^n u_r$ to be evaluated as follows.
- This method reduces the number of terms involved, allowing the series to be evaluated more easily.
- Partial fraction or algebra method are usually used to change the form of the addend, but other methods may be used if appropriate.
- Depending on the expression of $f(r + 1)$ above, the summation might,
 - **Converge**, if it gets closer and closer to certain value as n increases.
 - **Diverge**, if it approaches infinity as n gradually increases.
 - **Oscillate**, if its value keeps changing in a fixed interval as n changes.

Standard Formulae (MF19):

		Proof
1.	$\sum_{r=1}^n c = c + c + c + \dots + c = nc$	-
2.	$\sum_{r=1}^n r = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$	Apply the arithmetic series formulae.
3.	$\sum_{r=1}^n r^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$	Apply method of differences
4.	$\sum_{r=1}^n r^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4} = \left(\sum_{r=1}^n r\right)^2$	Can be shown using method of differences, but it is shown by using Mathematical Induction (Chapter 6)

Technique for solving problem:

- Apply the 4 formula above.
- Break up more complex sums using the three summation rules.
- Factor the expression whenever possible (don't multiply brackets unless no choice).

Proving:

(I) Use the Standard Formula to find the Sum of Certain Series**- Example 1:**

Prove that $\left(r + \frac{1}{2}\right)^3 - \left(r - \frac{1}{2}\right)^3 = 3r^2 + \frac{1}{4}$.

Use this result to prove that $\sum_{r=1}^n r^2 = \frac{1}{6}n(n+1)(2n+1)$.

Prove that $\sum_{n=1}^N (n^2 + 2n) = \frac{1}{6}N(N+1)(2N+7)$.

- **Example 2:**

Prove that $r^3 = \left[\frac{r(r+1)}{2} \right]^2 - \left[\frac{r(r-1)}{2} \right]^2$.

Use this result to prove that $\sum_{r=1}^n r^3 = \left[\frac{n(n+1)}{2} \right]^2$.

- **Example 3:**

Find the value of n for which $\sum_{r=1}^{2n} (48r - r^2) = 3 \sum_{r=1}^n (48r - r^2)$.

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- **Example 4:**

Prove that $\sum_{r=1}^n (2r - 1)(2r^2 - 2r + 1) = n^4$

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- **Example 5:**

Show that $\sum_{r=1}^n r(r+1)(r+2) = \frac{1}{4}n(n+1)(n+2)(n+3)$.

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- **Example 6:**

Prove that $(1^3 + 2^3 - 3^3) + (4^3 + 5^3 - 6^3) + \dots + [(3n - 2)^3 + (3n - 1)^3 - (3n)^3] = \frac{9}{4}n^2(3n^2 - 6n - 5)$.

(II) Use the Method of Differences**- Example 7:**

Prove that $\sum_{n=1}^N \frac{1}{\sqrt{n} + \sqrt{n+1}} = \sqrt{N}$.

Deduce that $\sum_{n=1}^N \frac{1}{\sqrt{n}} < 2\sqrt{N}$.

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- **Example 8:**

Find the sum of the series $\sum_{n=1}^N \frac{n^2+n+1}{n^2+n}$.

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- **Example 9:**

$$\text{Let } f(n) = \frac{4^{n+1}(n-1)}{3(n+2)}.$$

(i) Show that, for $n \geq 1$, $f(n) - f(n-1) = \frac{n^2 4^n}{(n+1)(n+2)}.$

(ii) Hence find, in terms of N , the sum of the series $\sum_{n=1}^N \frac{n^2 4^n}{(n+1)(n+2)}.$

- **Example 10:**

Find the sum to n terms of the series $1(1!) + 2(2!) + \dots + r(r!) + \dots$

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- **Example 11:**

Find the numbers A, B, C such that, for all values of r , $(r - 1)^2 = A(r + 2)(r + 1) + B(r + 1) + C$.

Hence, or otherwise, prove that $\sum_{r=1}^N (r - 1)^2 (r!) = (N - 2)[(N + 1)!] + 2$.

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- **Example 12:**

Let $a_n = e^{1-n} - e^{-n} + \frac{1}{n+1} - \frac{1}{n}$.

- (i) Find, in terms of N , an expression for S_N , where $S_N = a_1 + a_2 + \cdots + a_N$.
- (ii) Show that $S_N > 0$ for all $N \geq 1$.

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- **Example 13:**

Verify that $(n + 1)^2 n^2 (2n + 1) - n^2 (n - 1)^2 (2n - 1) = 10n^4 + 2n^2$.

Hence prove that $\sum_{n=1}^N n^4 = \frac{1}{30} N(N + 1)(2N + 1)(3N^2 + 3N - 1)$.

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- **Example 14:**

Express $\frac{r+4}{r(r+1)(r+2)}$ as partial fractions.

Prove that $\sum_{r=1}^n \frac{r+4}{2^r r(r+1)(r+2)} = p - \frac{q}{2^n(n+1)(n+2)}$, where p and q are constants to be determined.

- **Example 15 (PYQ M/J 2002):**

3 Given that

$$S_N = \sum_{n=1}^N (2n-1)^3,$$

show that

$$S_N = N^2(2N^2 - 1). \quad [4]$$

Hence find

$$\sum_{n=N+1}^{2N} (2n-1)^3$$

in terms of N , simplifying your answer. [3]

- **Example 16 (PYQ O/N 2002):**

1 Given that

$$u_n = e^{nx} - e^{(n+1)x},$$

find $\sum_{n=1}^N u_n$ in terms of N and x . [2]

Hence determine the set of values of x for which the infinite series

$$u_1 + u_2 + u_3 + \dots$$

is convergent and give the sum to infinity for cases where this exists. [3]

- **Example 17 (PYQ M/J 2003):**

- 3 Let $v_1, v_2, v_3, \dots$ be a sequence and let

$$u_n = nv_n - (n+1)v_{n+1},$$

for $n = 1, 2, 3, \dots$. Find $\sum_{n=1}^N u_n$. [2]

In each of the following cases determine whether the series $u_1 + u_2 + u_3 + \dots$ is convergent, and justify your conclusion. Give the sum to infinity where this exists.

(i) $v_n = n^{-\frac{1}{2}}$. [2]

(ii) $v_n = n^{-\frac{3}{2}}$. [2]

- **Example 18 (PYQ O/N 2004):**

5 Let

$$S_N = \sum_{n=1}^N (-1)^{n-1} n^3.$$

Find S_{2N} in terms of N , simplifying your answer as far as possible. [4]

Hence write down an expression for S_{2N+1} and find the limit, as $N \rightarrow \infty$, of $\frac{S_{2N+1}}{N^3}$. [3]

- **Example 19 (PYQ M/J 2007):**

1 Verify that

$$\frac{1}{n^2 + 1} - \frac{1}{(n + 1)^2 + 1} = \frac{2n + 1}{(n^2 + 1)(n^2 + 2n + 2)}. \quad [1]$$

Use the method of differences to show that, for all $N \geq 1$,

$$\sum_{n=1}^N \frac{2n + 1}{(n^2 + 1)(n^2 + 2n + 2)} < \frac{1}{2}. \quad [3]$$

Write down the value of

$$\sum_{n=1}^{\infty} \frac{2n + 1}{(n^2 + 1)(n^2 + 2n + 2)}. \quad [1]$$

- **Example 20 (PYQ M/J 2008):**

2 Given that

$$u_n = \ln\left(\frac{1+x^{n+1}}{1+x^n}\right),$$

where $x > -1$, find $\sum_{n=1}^N u_n$ in terms of N and x . [3]

Find the sum to infinity of the series

$$u_1 + u_2 + u_3 + \dots$$

when

(i) $-1 < x < 1$, [1]

(ii) $x = 1$. [1]



Example 21 (PYQ O/N 2009):

Prove by induction that

$$\sum_{n=1}^N n^3 = \frac{1}{4}N^2(N+1)^2. \quad [5]$$

Use this result, together with the formula for $\sum_{n=1}^N n^2$, to show that

$$\sum_{n=1}^N (20n^3 + 36n^2) = N(N+1)(N+3)(5N+2). \quad [3]$$

Let

$$S_N = \sum_{n=1}^N (20n^3 + 36n^2 + \mu n).$$

Find the value of the constant μ such that S_N is of the form $N^2(N+1)(aN+b)$, where the constants a and b are to be determined. [3]

Show that, for this value of μ ,

$$5 + \frac{22}{N} < N^{-4}S_N < 5 + \frac{23}{N},$$

for all $N \geq 18$. [3]

Solution:

Base case:

$n = 1$

$$\text{LHS: } \sum_{r=1}^1 r^3 = 1 \quad \text{RHS: } \frac{1}{4}(1)(2)^2 = 1$$

LHS = RHS, hence the statement is true for $n = 1$.

Inductive step: Assume the statement is true for some integer k : $\sum_{r=1}^k r^3 = \frac{1}{4}k^2(k+1)^2$.

What we want to prove: when $n = k + 1$: $\sum_{r=1}^{k+1} r^3 = \frac{1}{4}(k+1)^2(k+2)^2$

$$\begin{aligned} \sum_{r=1}^{k+1} r^3 &= \sum_{r=1}^k r^3 + (k+1)^3 \\ &= \frac{1}{4}k^2(k+1)^2 + (k+1)^3 \\ &= \frac{1}{4}(k+1)^2[k^2 + 4(k+1)] = \frac{1}{4}(k+1)^2(k+2)^2 \end{aligned}$$

Hence the statement is true for $n = k + 1$. By principle of mathematical induction,

$$\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n+1)^2 \text{ for all positive integers } n.$$



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- **Example 22 (PYQ M/J 2010/13):**

- 2 By considering the identity

$$\cos[(2n-1)\alpha] - \cos[(2n+1)\alpha] \equiv 2 \sin \alpha \sin 2n\alpha,$$

show that if α is not an integer multiple of π then

$$\sum_{n=1}^N \sin(2n\alpha) = \frac{1}{2} \cot \alpha - \frac{1}{2} \operatorname{cosec} \alpha \cos[(2N+1)\alpha]. \quad [4]$$

Deduce that the infinite series

$$\sum_{n=1}^{\infty} \sin\left(\frac{2}{3}n\pi\right)$$

does not converge.

[1]

- **Example 23 (PYQ O/N 2011/13/1):**

Verify that $\frac{1}{n^2} - \frac{1}{(n+1)^2} = \frac{2n+1}{n^2(n+1)^2}$. [1]

Let $S_N = \sum_{r=1}^N \frac{2r+1}{r^2(r+1)^2}$. Express S_N in terms of N . [2]

Let $S = \lim_{N \rightarrow \infty} S_N$. Find the least value of N such that $S - S_N < 10^{-16}$. [3]